

Bad Idea or Good Prediction? Comparing VLM and Human Anticipatory Judgment

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Abstract

Anticipating outcomes of everyday scenarios is crucial for robots operating in human environments. This study evaluates the anticipatory reasoning capabilities of state-of-the-art Vision Language Models (VLMs) by showing them videos of human and robot scenarios with outcomes removed, then asking them to predict whether the situation will end well or poorly. We tested multiple VLMs, including closed-source, open-source, and VLM-LLM combinations, using various prompts and compared their predictions against both true outcomes and judgments from 29 human participants. The best-performing VLM configuration achieved 70.0% accuracy in predicting true outcomes, outperforming the average individual human ($62.1\% \pm 6.2\%$), while other configurations ranged from 43.3% to 60.0%. Alignment with individual human judgments ranged from 44.4% to 69.7%. Prompt engineering significantly impacted performance, with variations up to 6.7% within the same model. Notably, VLMs struggled to predict outcomes by analyzing human facial reactions, achieving only 48% accuracy, suggesting limitations in leveraging social cues. These preliminary findings indicate that while some VLMs show promise for anticipatory reasoning, their performance is highly sensitive to model selection and prompt design, with current limitations in social intelligence that warrant further investigation for human-robot interaction applications.

Keywords

social competence; robot error; anticipation; human-AI collaboration

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1 Introduction

TODO: add more refs to replace related work

Vision-Language Models (VLMs) are increasingly being deployed in human-robot interaction systems, with applications ranging from task planning and navigation to manipulation and multimodal perception [2?]. These models enable robots to interpret contextual cues and execute tasks aligned with human intentions by processing and correlating visual data from cameras with linguistic inputs.

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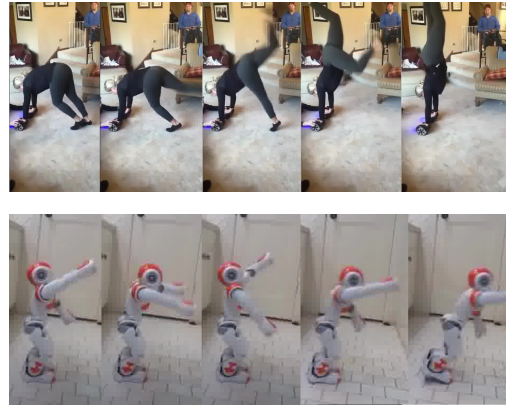


Figure 1: In Parreira et al. [3], human participants were shown videos displaying a variety of scenarios featuring robots and humans. Videos ended before the outcome was displayed, and participants were asked to anticipate the outcome of each video. The same video dataset is used in this study.

As robots become more prevalent in shared human spaces, developing social intelligence – the ability to perceive, understand, and appropriately respond to social contexts – becomes critical for safe and effective interaction [].

Recent HRI research has increasingly leveraged VLMs for diverse applications including scene understanding, human intention recognition, and collaborative task execution. For example, in Sasabuchi et al. [4], a combination of VLMs for scene understanding and LLMs for judgment performed well on a task of deciding when to initiate interaction. However, most applications focus on task execution and scene understanding rather than *anticipatory* reasoning: the ability to predict whether a situation will end well or poorly before the outcome occurs. This gap is particularly critical for safety-sensitive scenarios where robots must proactively prevent errors rather than reactively respond to them. While anticipatory capabilities are fundamental to human social cognition, enabling us to constantly evaluate action consequences through contextual observation, whether current VLMs possess comparable predictive abilities remains unexplored.

This paper presents an **exploratory evaluation of VLM anticipatory reasoning capabilities** across multiple dimensions. Drawing on the "Bad Idea?" human study [3], we test closed-source, open-source, and VLM-LLM combination models on outcome prediction tasks, comparing their performance against both ground truth outcomes and human judgments from 29 participants. Our investigation reveals several preliminary findings: (1) some VLMs can exceed average human performance on outcome prediction, though with high sensitivity to model and prompt selection; (2) open-source models substantially underperformed closed-source alternatives, with some exhibiting severe prediction bias; and (3)

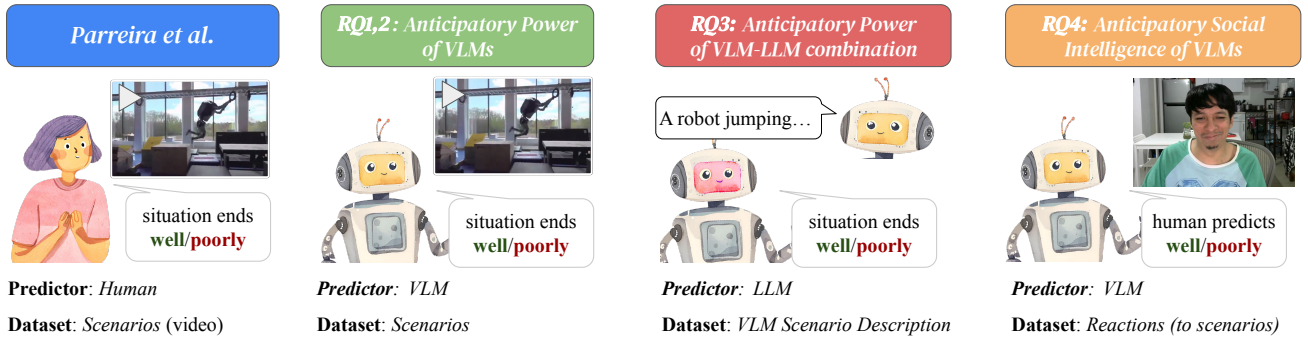


Figure 2: rqs and datasets summary

in a novel application analyzing human facial reactions to predict outcomes, VLMs struggled substantially, suggesting fundamental limitations in leveraging social cues. While exploratory in scope, these findings provide initial insights into both the promise and current limitations of VLMs for anticipatory reasoning in HRI contexts, highlighting critical considerations for deploying these models in safety-critical applications.

2 Methods

We describe the dataset, models and methods used in this study, which aims to test the *anticipatory judgement* (ability to predict outcomes based on anticipatory context) of VLMs. We investigate this across two datasets: the **scenario dataset** includes a set of videos which display a scenario, but the outcome is cut off; the **reactions dataset** is composed of human participant reactions to those videos, and respective outcome prediction (Figure 2).

We explore the following research questions:

- **RQ1:** What is the anticipatory judgement of closed-source VLMs (i.e., predicting outcomes on the *scenario dataset*)?
- **RQ2:** What is the anticipatory judgement of open-source VLMs (i.e., predicting outcomes on the *scenario dataset*)?
- **RQ3:** What is the anticipatory judgement given a VLM-LLM combination?
- **RQ4:** What is the anticipatory social intelligence of closed-source VLMs (i.e., predicting outcomes based on the *reactions dataset*)?

2.1 Dataset

User study: In Parreira et al. [3], the authors conducted an online study to collect anticipatory human reactions to stimulus videos through Prolific, in line with previous work [1]. Participants were shown a series of scenarios through short videos. The protocol included a “warm-up” round of 3 videos, followed by a main round of data collection (30 videos). Each video was shown twice: first, a shorter version of the video, which stopped before the video action was fully resolved (e.g., someone swinging from a rope, approaching a tree branch). After watching this video, participants were asked “You think this situation ends...” with the options “well” or “poorly”. After this, participants were able watch a longer version of the

video, featuring the resolution of the video action. Full details on the study protocol and ethics can be found in the original paper.

Stimulus dataset: A set of 30 videos, which includes scenarios where humans and robots are featured and there are good and bad outcomes, was used (Figure 1). The full list can be found on Parreira et al. [3] study repository ¹.

Participants: TOTAL REACTIONS FROM TOTAL PARTICIPANTS... also look for what is the shortlist of participants to test performance on those

2.2 Prompts

We tested multiple prompts to evaluate VLMs’ anticipatory reasoning capabilities. For direct outcome prediction on the *scenario dataset*, we employed two primary prompts that directly query the model’s judgment:

Prompt A (Direct): “Given the scenario shown on the video, you think this situation ends well or poorly? (Use only one word to answer)”

Prompt B (Human-like): “Given the scenario shown on the video, you think this situation ends well or poorly as if you are a human watching the video? (Use only one word to answer)”

For RQ3, which explores VLM-LLM combinations, we additionally employed descriptive prompts to generate text representations of the scenarios:

Prompt C (Descriptive): “Output a sentence describing what’s going on in the video, keeping only key elements of the scenario shown that would allow me to predict the outcome of the situation.”

Prompt D.1 (Predictive): “Given the scenario described, you think this situation ends well or poorly? (Use only one word to answer)”

Prompt D.2: “Analyze this scenario and predict the outcome. Respond with exactly one word: ‘well’ if the situation has a positive resolution, or ‘poorly’ if it has a negative resolution.”

For RQ4, which evaluates anticipatory social judgment based on human reactions, we adapted the prompts to focus on interpreting facial expressions and reactions rather than the scenario itself.

¹<https://github.com/IRL-CT/badidea>

Prompt E (Reaction-based): “Given the human reaction shown in the image, do you think the situation observed by that human ends well or poorly? (Use only one word – well or poorly – to answer)”

2.3 Models Tested

We evaluated models across multiple research questions, selecting both closed-source and open-source VLMs to assess the current state of anticipatory reasoning capabilities. We used Ollama² to cue local models. Since many VLMs are designed for image input rather than video, we employed frame extraction strategies detailed below.

2.3.1 RQ1: Closed-source VLMs (Scenario Dataset). For evaluating anticipatory judgement of commercial VLMs on the scenario dataset, we tested three state-of-the-art closed-source models on scenario videos: **GPT-4o**, **Gemini 2.0 Flash**, **Qwen-2.5-vl-72b**

2.3.2 RQ2: Open-source VLMs (Scenario Dataset). To evaluate the anticipatory power of publicly available models, we tested eight open-source VLMs that can be deployed locally: DeepSeek-OCR, LLaVA 3.2 Vision, LLaVA, Qwen3-vl, Gemma 3 (SEE number of parameter), Mistral Small 3.2 (Pixtral), LLaVA-LLaMA3.

These models process images, not video. Thus, we extracted frames from each video (15 fps) and obtained predictions for each. The final prediction was determined either by the **mode** (most frequent prediction) across all frames or by the **last** frame prediction. We tested these models only on *prompt A*, following observations of higher performance in outcome prediction from RQ1.

2.3.3 RQ3: VLM-LLM Combinations. To explore whether combining vision-language understanding with language reasoning improves performance, we tested a two-stage pipeline: **VLM Stage:** Use VLMs (Gemini 2.0 Flash with Prompt C) to generate text descriptions of video scenarios; **LLM Stage:** feed generated descriptions to large language models for outcome prediction (Gemini 2.0 Flash).

This approach tests whether explicit scene understanding through text generation enhances anticipatory reasoning compared to end-to-end visual prediction.

2.3.4 RQ4: Anticipatory Social Intelligence (Reactions Dataset). For evaluating VLMs’ ability to interpret human anticipatory reactions, we employed the open-source models from RQ2 on the *reactions dataset*. These models analyzed recordings of participant facial expressions and reactions while watching scenario videos, with the task of predicting their anticipated outcome.

For the reactions dataset, we tested two temporal window configurations:

- **1-second window:** Extracted frames at 2 fps from the final 1 second before video end
- **3-second window:** Extracted frames at 2 fps from the final 3 seconds before video end

The final prediction was determined by the **mode** across all extracted frames within the window or the prediction from the **last** frame in the sequence.

²<https://ollama.com/>

2.4 Evaluation Metrics

We evaluated model performance using standard binary classification metrics: accuracy, precision, recall, and f1-score, as well as poorly prectictio ratio (ratio of poorly: total predicted outcomes).

For comparison with human performance, we computed two types of alignment:

- (1) **Ground Truth Alignment:** Model predictions compared against actual video outcomes
- (2) **Human Alignment:** Model predictions compared against individual human participant judgments (reported as mean $\pm$ standard deviation across all participants)

3 Results

INTRO SENTENCE

RQ1: Anticipatory power of closed-source VLMs We evaluated three closed-source VLMs using two prompt variants on the scenario dataset. Table 1 presents model performance against ground truth outcomes, while Table 2 shows alignment with individual human predictions.

Table 1: Closed-source VLM performance compared to ground truth outcomes, for Prompts A and B. Human baseline shown as $M \pm SD$ across 29 participants. PR: ratio of "Poorly" predicted outcomes to total predictions.

Model	Acc	Prec	Rec	F1	PR
GPT-4o (A)	0.433	0.375	0.462	0.414	0.467
GPT-4o (B)	0.467	0.400	0.462	0.429	0.500
Qwen (A)	0.500	0.450	0.692	0.545	0.333
Qwen (B)	0.533	0.476	0.769	0.588	0.300
Gemini (A)	0.700	0.625	0.769	0.690	0.467
Gemini (B)	0.633	0.562	0.692	0.621	0.467
Human Average	0.621 $\pm$ 0.062	0.575 $\pm$ 0.086	0.599 $\pm$ 0.091	0.579 $\pm$ 0.056	—

Table 2: Closed-source VLM alignment with individual human predictions, for Prompts A and B. Values shown as $M \pm SD$ across comparisons with 29 participants.

Model	Acc	Prec	Rec	F1
GPT-4o (A)	0.489 $\pm$ 0.076	0.457 $\pm$ 0.126	0.522 $\pm$ 0.091	0.482 $\pm$ 0.105
GPT-4o (B)	0.444 $\pm$ 0.072	0.408 $\pm$ 0.102	0.440 $\pm$ 0.074	0.418 $\pm$ 0.082
Qwen (A)	0.605 $\pm$ 0.087	0.553 $\pm$ 0.125	0.794 $\pm$ 0.071	0.644 $\pm$ 0.101
Qwen (B)	0.639 $\pm$ 0.087	0.574 $\pm$ 0.119	0.871 $\pm$ 0.077	0.684 $\pm$ 0.095
Gemini (A)	0.697 $\pm$ 0.070	0.651 $\pm$ 0.129	0.755 $\pm$ 0.076	0.691 $\pm$ 0.087
Gemini (B)	0.692 $\pm$ 0.069	0.647 $\pm$ 0.130	0.750 $\pm$ 0.074	0.686 $\pm$ 0.087

Key findings from RQ1 include: (1) Gemini 2.0 Flash outperformed both other models and the average human on predicting true outcomes, (2) prompt engineering significantly impacted performance, with differences up to 6.7 percentage points within the same model (Gemini Prompt A vs. Prompt B), and (3) models showing higher alignment with humans did not necessarily achieve higher accuracy on true outcomes, suggesting humans and models may share similar biases.

RQ2 Anticipatory power of open-source VLMs We evaluated eight open-source VLMs from Ollama on the scenario dataset using Prompt A. Since these models process images rather than video, we tested two aggregation methods: mode (most frequent prediction across all frames) and last frame prediction. Table 3 presents the best-performing method for each model against ground truth outcomes, while Table 4 shows alignment with individual human predictions.

Table 3: Open-source VLM performance compared to ground truth outcomes. For each model, the best-performing aggregation method (MODE or LAST) is shown. Human baseline shown as $M \pm SD$ across 29 participants. PR: ratio of "Poorly" predicted outcomes to total predictions.

Model	Acc	Prec	Rec	F1	PR	Method
LLaMA 3.2 Vision	0.433	0.433	1.000	0.605	1.000	MODE
LLaVA-LLaMA 3	0.533	0.400	0.154	0.222	0.167	MODE
DeepSeek-OCR	0.500	0.250	0.077	0.118	0.133	MODE
Gemma 3	0.433	0.433	1.000	0.605	1.000	MODE
Qwen 2.5	0.533	0.462	0.462	0.462	0.433	LAST
Mistral Small 3.2	0.433	0.433	1.000	0.605	1.000	MODE
Qwen 3	0.467	0.286	0.154	0.200	0.233	MODE
LLaVA	0.367	0.375	0.692	0.486	0.800	LAST
Human	0.621 ± 0.062	0.575 ± 0.086	0.599 ± 0.091	0.579 ± 0.056	—	—

Table 4: Open-source VLM alignment with individual human predictions. For each model, results using the same method as in Table 3 are shown. Values shown as $M \pm SD$ across comparisons with 29 individual participants.

Model	Acc	Prec	Rec	F1	Method
LLaMA 3.2 Vision	0.465 ± 0.097	0.465	1.000	0.628	MODE
LLaVA-LLaMA 3	0.510 ± 0.085	0.424	0.151	0.220	MODE
DeepSeek-OCR	0.584 ± 0.080	0.681	0.200	0.306	MODE
Gemma 3	0.465 ± 0.097	0.465	1.000	0.628	MODE
Mistral Small 3.2	0.465 ± 0.097	0.465	1.000	0.628	MODE
Qwen 3	0.545 ± 0.065	0.521	0.258	0.341	MODE
LLaVA	0.494 ± 0.068	0.474	0.816	0.593	LAST

Key findings from RQ2 include: (1) open-source models underperformed closed-source alternatives, with the best open-source models (LLaVA-LLaMA 3 and Qwen 2.5 at 53.3%) achieving 17 percentage points lower accuracy than the best closed-source model (Gemini at 70.0%), (2) several open-source models exhibited severe prediction bias, predicting "poorly" for nearly all scenarios, suggesting difficulty in discriminating between different outcome types, (3) frame aggregation method (MODE vs LAST) impacted performance, with most models benefiting from MODE aggregation but Qwen 2.5 and LLaVA performing better with LAST frame prediction, and (4) even the best-performing open-source models remained below human-level performance, highlighting the gap between open and closed-source VLM capabilities for anticipatory reasoning.

RQ3: Anticipatory power of VLM-LLM combinations We evaluated a two-stage VLM-LLM pipeline where Gemini 2.0 Flash generates text descriptions of scenarios, which are then processed for outcome prediction. Since the VLM stage processes video frames, we tested MODE and LAST aggregation methods. Table 5 presents performance against ground truth outcomes, while Table 6 shows alignment with individual human predictions.

Table 5: VLM-LLM combination performance compared to ground truth outcomes. LAST frame aggregation performed best for both prompt types. Human baseline shown as $M \pm SD$ across 29 participants. PR: ratio of "Poorly" predicted outcomes to total predictions.

Model	Acc	Prec	Rec	F1	PR
Gemini 2.0 (D.1)	0.500	0.458	0.846	0.595	0.800
Gemini 2.0 (D.2)	0.600	0.545	0.462	0.500	0.367
Human	0.621 ± 0.062	0.575 ± 0.086	0.599 ± 0.091	0.579 ± 0.056	—

Table 6: VLM-LLM combination alignment with individual human predictions. Values shown as $M \pm SD$ across comparisons with 29 participants.

Model	Acc	Prec	Rec	F1
Gemini 2.0 (Desc)	0.521 ± 0.071	0.491	0.850	0.616
Gemini 2.0 (Direct)	0.563 ± 0.067	0.538	0.425	0.469

Key findings from RQ3 include: (1) the two-stage VLM-LLM approach achieved competitive performance, with the direct prompt configuration reaching 60.0% accuracy compared to 70.0% for the best end-to-end VLM (Gemini with Prompt A in RQ1), (2) generating explicit text descriptions before prediction (descriptive prompt) did not improve performance over direct outcome queries, suggesting that intermediate text representation may lose critical visual information, (3) LAST frame aggregation consistently outperformed MODE for both prompt types in the VLM-LLM pipeline, contrary to most open-source models in RQ2, and (4) the VLM-LLM combination remained below human performance, indicating that decomposing the task into description generation and outcome prediction does not bridge the anticipatory reasoning gap.

RQ4: Anticipatory Social Intelligence We evaluated open-source VLMs' ability to predict outcomes by analyzing human facial reactions while watching scenario videos. Models analyzed frames extracted from 1-second or 3-second windows before participants made their predictions. Table 7 shows model alignment *with human predictions*—the model prediction is mapped to what that participant predicted, not the ground truth outcome.

Key findings from RQ4 include: (1) VLMs performed poorly at predicting outcomes from human reactions, with most models achieving accuracy below 50%, (2) many models exhibited extreme prediction bias (poorly ratio near 0.0 or 1.0), suggesting they struggled to extract meaningful discriminative features from facial expressions,

Table 7: Open-source VLM performance on predicting outcomes from human reactions. Performance represents alignment with human predictions (not ground truth). PR: ratio of "Poorly" predicted outcomes to total predictions.

Model	Window	Acc	Prec	Rec	F1	PR
DeepSeek-OCR	1s	0.535	0.000	0.000	0.000	0.000
DeepSeek-OCR	3s	0.519	0.000	0.000	0.000	0.000
LLaVA	3s	0.481	0.481	1.000	0.649	1.000
Qwen 3	1s	0.479	0.455	0.617	0.524	0.630
Qwen 3	3s	0.479	0.461	0.502	0.481	0.523
LLaMA 3.2 Vision	3s	0.478	0.473	0.842	0.606	0.848
LLaMA 3.2 Vision	1s	0.465	0.465	0.995	0.634	0.996
LLaVA	1s	0.464	0.464	0.998	0.633	0.999
Gemma 3	3s	0.467	0.473	0.966	0.635	0.982
Mistral Small 3.2	1s	0.458	0.456	0.853	0.594	0.871
Gemma 3	1s	0.453	0.458	0.963	0.621	0.977
LLaVA-LLaMA 3	1s	0.446	0.436	0.652	0.522	0.695
Mistral Small 3.2	3s	0.436	0.453	0.835	0.588	0.886
LLaVA-LLaMA 3	3s	0.434	0.434	0.586	0.499	0.649

(3) no significant performance difference between 1-second and 3-second temporal windows, indicating that temporal context length does not substantially aid reaction interpretation, and (4) the task of interpreting human anticipatory reactions appears substantially more challenging for current VLMs than direct scenario analysis, indicating fundamental limitations in social intelligence and subtle emotional cue recognition. The best discriminative model (Qwen 3) achieved only 48% accuracy compared to human self-agreement of 62%, demonstrating a significant gap in VLMs' ability to leverage human social cues for outcome prediction.

4 Discussion

TODO: add references that align or not with observations test description of facial expressions for outcome prediction llms

This exploratory study provides initial evidence that some state-of-the-art VLMs can perform anticipatory reasoning tasks, with Gemini 2.0 Flash achieving 70.0% accuracy on our scenario dataset (exceeding the average human performance of 62.1%). However, these preliminary results suggest considerable fragility in this capability, with performance varying by up to 23.4 percentage points across different models and prompts. Whether this represents genuine anticipatory reasoning or pattern matching on visual cues remains an open question requiring further investigation.

Our limited comparison between closed-source and open-source VLMs hints at substantial performance differences (70.0% vs 53.3% at best), though the small scale of this study prevents strong conclusions. Several open-source models exhibited severe prediction bias, suggesting possible limitations in discriminating between outcome types. However, given the exploratory nature of this work and the rapid evolution of open-source models, these observations should be interpreted cautiously as snapshots rather than definitive assessments.

An unexpected finding emerged from our frame aggregation analysis: while most open-source models appeared to benefit from

temporal consensus (MODE), VLM-LLM pipelines favored last-frame predictions. This pattern suggests potentially different processing characteristics, though alternative explanations—including dataset-specific artifacts or prompt interactions—cannot be ruled out with our limited sample. The VLM-LLM approach achieved competitive performance (60.0% accuracy at best), but whether decomposing the task into description and prediction stages offers genuine advantages remains unclear.

Perhaps the most striking preliminary observation concerns VLMs' apparent difficulty with interpreting human anticipatory reactions (RQ4), where models achieved only 48% accuracy compared to 62% human self-agreement. This suggests current VLMs may struggle to leverage social cues for outcome prediction, though our small dataset and limited model selection constrain firm conclusions. If replicated in larger studies, this finding could have important implications for HRI applications requiring interpretation of human emotional states and anticipatory behaviors.

Limitations. This work is explicitly exploratory, and several significant limitations constrain interpretation. Our evaluation uses only 30 videos—far too small to establish generalizable conclusions about VLM capabilities. The human baseline comes from a specific online population and may not represent broader human performance. We tested a limited selection of models (3 closed-source, 8 open-source) and only a handful of prompt formulations; different model versions or systematic prompt engineering might substantially change results. Our frame aggregation analysis examined only two simple strategies (MODE and LAST); more sophisticated approaches could yield different patterns. The rapid pace of VLM development means our findings may already be outdated. Finally, our binary outcome framing (well/poorly) oversimplifies real-world anticipatory reasoning. These results should be viewed as preliminary observations that motivate—rather than answer—questions about VLM anticipatory capabilities.

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